Interim Information Design And Buyer-Optimal Pricing*

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Abstract

We study efficient and buyer-surplus maximizing information policies in a bilateral trade setting where the buyer is initially privately imperfectly informed about his willingness to pay. We identify a canonical class of demand functions that can be implemented by information disclosures that are targeted based on the buyer's initial private information. As an application we show that providing more information to the buyer can lead to higher market prices and a lower trade probability without affecting the buyer or seller surplus.

Introduction

We study the information design problem in a single-item bilateral trade environment where private disclosures to the buyer can impact her willingness to pay. The overall policy of such disclosures determines the demand faced by the seller and shapes prices, buyer surplus and profits. See for example Roesler and Szentes (2017).

We analyze information design from an *interim* perspective. The buyer begins

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with some initial, partial, private information about her value for the object. This allows for disclosures that are targeted based on the buyer's private information but constrains the overall information policy to be no less informative than the initially partially informed buyer. We ask how market outcomes, like the probability of trade, prices, and welfare, are affected by interim information disclosures. In particular, we study how the designer's information policy can shape an *interim demand* into a *market demand* which determines the price and welfare.

We entertain three interpretations of interim information design. First, the information policies we derive can be thought of as a "first-best" benchmark relative to which any decentralized policy subject to incentive constraints would naturally be compared. Studies of incentive-constrained interim information design include Guo, Li and Shi (2025), Bergemann, Bonatti and Smolin (2018), Ely (2025), and Chopra and Ely (2026). Second, we may think of the buyer-optimal policies as simply describing how the buyer herself would optimally acquire information were she able to commit to a type-dependent experiment, a direct generalization of Roesler and Szentes (2017) to the case of an initially privately informed buyer. Finally, and as our main applied motivation, our model could represent transactions that take place through a digital market platform which collects and discloses information to the participants, as in Bergemann, Bonatti and Smolin (2018). The buyer's private information came from previous transactions, tracked by the platform.

Our main applied result relates prices to information and welfare with important implications for regulatory oversight. Higher prices, say on the digital platform, can be entirely welfare neutral and arise solely because the platform provides the buyer with better information. In particular, improved information allows buyers to make better purchase decisions and this value can offset, even dominate the welfare costs of higher prices. More generally, we provide a comparative static relationship between prices and information. Fixing some Pareto efficient buyer-seller surplus pair, as the buyer becomes better informed, prices necessarily increase.

Our theoretical contribution is to characterize all possible buyer-seller surplus pairs for any given distribution of initial private information. A result central to our analysis is the identification of a new class of canonical demands that are sufficient to generate all possible efficient buyer-seller surplus pairs. These demands resemble the iso-elastic demands that play a key role in Roesler and Szentes (2017), but have gaps and atoms resulting from the constraint imposed by the buyer's initial private information. We show that any Pareto efficient equilibrium demand can be transformed into a demand in this family through some information policy without affecting the welfare or price.

Roesler and Szentes (2017) characterize all possible equilibrium buyer-seller surplus pairs in a framework in which the buyer is initially uninformed (or equivalently the seller and buyer are initially symmetrically informed). They show that buyer-surplus maximizing information policies are efficient and entail trade with probability 1. As trade always occurs in an efficient equilibrium, the total buyer-seller surplus equals the buyer's prior expectation of the item's valuation, and the seller's surplus equals the price charged. As a result, in Roesler and Szentes (2017), there is a one-to-one relationship between the market price and the buyer-seller surplus pair for any efficient outcome.

In our model, different market prices can support any given Pareto efficient equilibrium buyer-seller surplus pair. Moreover, trade can occur with probability less than 1 for Pareto efficient outcomes due to the buyer's initial private information, resulting in extra flexibility between equilibrium welfare and prices.

Related Literature

Roesler and Szentes (2017) initiated the study of buyer-optimal information design in a bilateral-trade context. Condorelli and Szentes (2025) analyzed a similar problem in which the distribution of buyer's valuations comes from a covert and costless investment decision rather than information acquisition. Park (2025) adapted the Roesler and Szentes (2017) framework to a setting in which a collection of buyers cooperatively purchase a public good. All of these papers study the design problem at the *ex ante* stage before the buyer(s) have any private information. Bergemann, Brooks and Morris (2015) study the welfare possibilities when information is provided to an initially uninformed seller about buyers' valuations.

In independent and concurrent research Ennuschat (2025) looks at a variation of the interim problem we study. In Ennuschat (2025) the underlying willingness to pay can take a continuum of values and the methodological approach is correspondingly different. Bergemann and Morris (2016) formulated the information design problem in which an initial information structure is held fixed and the designer can provide additional information, see also Bergemann and Morris (2019).

Our paper is also related to work on information design with privately informed agents initiated by Kolotilin, Mylovanov, Zapechelnyuk and Li (2017). Bergemann, Bonatti and Smolin (2018) study a monopolistic intermediary that screens heterogeneous buyers through a menu of information products. Guo, Li and Shi (2025) study a bilateral-trade mechanism that jointly screens a privately informed buyer through sequential pricing and type-dependent disclosure. Chopra (2025), Chopra and Ely (2026), and Ely (2025) are further applications in which intermediaries screen privately informed agents through menus of tests or information policies. Our analysis contributes to that literature by providing tools to characterize a first-best benchmark wherein information can be conditioned directly on the buyer's interim type.

Model

We study a bilateral trade model with a buyer (he) and a monopolist seller (she). The seller offers a single item, produced at a cost c. The buyer's underlying value for the product is binary and represented by $\theta \in \{0, 1\}$. The seller's cost c is assumed to satisfy $0 \leq c < 1$, making trade efficient when $\theta = 1$. We refer to the benchmark case $c = 0$ as the no-gap case.

The buyer has partial private information about his valuation, represented by his prior belief $\mu \in [0, 1]$. We refer to the buyer's private belief μ as his type. The buyer's type is his interim expected valuation for the item.

There is a commonly known distribution $F \in \Delta([0, 1])$ from which types are drawn. The distribution F is the interim demand that the seller faces. The designer announces an information policy, denoted by $\rho(\cdot) = (\rho_0(\cdot), \rho_1(\cdot))$, where for each μ

$$\rho_\theta(\mu) \in \Delta(\mathcal{M})$$

is the distribution over signals that the buyer observes conditional on the true value of the object being θ. Here $\rho(\mu)$ represents a statistical experiment about θ whose realization is in some arbitrary set $\mathcal{M}$ and is only observed by the type μ buyer.

A type μ buyer observes the realization of $\rho(\mu)$ and updates his prior expected valuation μ to a posterior expected valuation ν as per Bayes rule. As the buyer's expected valuation dictates his decision to trade, each experiment $\rho(\mu)$ can be identified with the distribution of posterior valuations it induces for type μ buyer. Let $\eta(\mu) \in \Delta([0, 1])$ represent this distribution of posterior expected valuations. In particular, it is without loss to identify an information policy with a mapping from types to a distribution of posterior expected valuations, represented by $\eta : [0,1] \to \Delta([0,1])$.

In response to the announced information policy, the seller forms a posterior distribution (*market demand*) G over the buyer's posterior expected valuation and offers a price p to the buyer. For any given information policy with corresponding η the market demand G is such that for all $\nu \in [0,1]$

$$G(\nu) = \mathbf{E}_F \left[\, \eta(\mu)(\, [0, \, \nu] \,) \, \right]$$

In other words $G(\nu)$ is the total probability under F and η that the buyer obtains posterior expected valuation less than or equal to ν.

Generally, more informative information policies generate more dispersed market demand. When the state space is binary, the converse also holds: more dispersed demands correspond to a better-informed buyer. Formally, by Theorem 4.7 in Elton and Hill (1992), the designer can induce a distribution G using some information policy $\rho(\cdot)$ if and only if G is a mean preserving spread of F, written $G \succeq_{mps} F$. The relation $G \succeq_{mps} F$ holds if and only if for every real valued continuous convex function φ we have the following

$$\int_0^1 \varphi(s) dG(s) \geq \int_0^1 \varphi(s) dF(s)$$

By Theorem 3.A.1 in Shaked and Shanthikumar (2007), the above is equivalent to the following

$$\int_0^p G(s) ds \geq \int_0^p F(s) ds \;\; \text{for all} \;\; p \in [0,1]$$

where the above holds with equality at $p = 1$.

We will refer to such a distribution G as a *feasible distribution* (or *feasible demand*). In general there can be different information policies that lead to the same market demand. The equilibrium outcomes depend on the information policy only through the induced market demand, as such we will work with the demand G instead of the underlying information policy.

When the state space is binary (and bounded), for some feasible demands G_1 and G_2, the relationship $G_1 \succeq_{mps} G_2$ holds if and only if the demand G_1 is generated by a more informative information policy than the information policy that induces G_2. In particular, we can obtain the demand G_1 from an interim belief G_2 through targeted information disclosure. Throughout the analysis, we utilize this relationship between feasible market demand and the information policy chosen by the designer. This relationship breaks down with more than two states (see, for example, Elton and Hill (1992)); in particular, a more dispersed feasible demand cannot necessarily be recovered from a less dispersed feasible demand through targeted disclosure.

Feasible Demands

As pointed out in the previous section, given a prior F, the set of feasible seller's beliefs about the distribution of buyer's valuations (type) is given by

$$\mathcal{G}(F) := \left\{ G : [0,1] \to [0,1] \mid G \succeq_{mps} F,\ G \text{ is a cdf } \right\}$$

In our interim setting, the buyer has initial private information and the designer can provide information about the value θ type-by-type. For each buyer type μ, there corresponds a distribution of posterior expected valuations $\eta(\mu)$ resulting from the information provided by the designer. The market demand faced by the seller is the average across types μ of these type-contingent posterior valuations.

By contrast, when the buyer and seller are initially symmetrically uninformed as in Roesler and Szentes (2017), the set of feasible market demands is given by

$$\mathcal{G}_{\text{UI}}(F) := \left\{ G : [0,1] \to [0,1] \mid F \succeq_{mps} G,\ G \text{ is a cdf } \right\}$$

Here is a way to understand the difference. In both models F is the seller's prior distribution over the buyer's valuation. In Roesler and Szentes (2017) the buyer begins with no information so F is also the buyer's prior distribution over her own valuations and therefore with probability 1 her interim willingness to pay is $\mathbf{E}F$. And in Roesler and Szentes (2017), the most the buyer could learn with full information is the realization of F. Thus in Roesler and Szentes (2017), the set of market demands is the set of mean-preserving *contractions* of F. By contrast in our framework the buyer begins already knowing the realization of F and therefore its realization represents the *least* the buyer could learn. The set of market demands is the set of mean-preserving *spreads* of F.

Equilibrium Outcomes

We will assume that the seller can dictate the terms of trade between the buyer and seller. Thus, we require that the buyer trades if and only if his posterior willingness to pay weakly exceeds p.[1]

The seller is a monopolist and her profit-maximizing price against a market demand G and a cost c is

$$p \in \text{argmax}_{s\in[0,1]}\ (s-c)\cdot(\ 1-G(s)\ +\Delta(G,s))$$

Where $\Delta(G,s)$ represents the size of the atom at s for the distribution G.[2]

$$\Delta(G,s) := \lim_{x\downarrow s} G(x) - \lim_{x\uparrow s} G(x)$$

An *equilibrium profile* is a tuple (G,p) such that 1) $G \in \mathcal{G}(F)$, 2) $p \in \text{argmax}_{s\in[0,1]}\ (s-c)\cdot(\ 1-G(s)\ +\Delta(G,s))$.
Each equilibrium profile has a corresponding *outcome* (κ,π) where κ is the buyer surplus and π is the seller surplus given by

$$\kappa := \int_p^1 (s-p)dG(s)$$

$$\pi := (p-c)\cdot(1-G(p)+\Delta(G,p))$$

Note that for any equilibrium profile (G,p) condition 2) requires that the market price p is weakly greater than the cost of production c and strictly greater whenever the demand G is not a degenerate point mass at c.

Pareto Efficient Demands

We start by describing necessary conditions under which an equilibrium profile is Pareto efficient.[3] If a feasible market demand G supports buyer valuation strictly between 0 (the lowest valuation) and the market price p then the designer can provide additional information targeted to these buyer valuations in a way that strictly increases both the buyer and seller surplus. The following lemma formalizes this statement.

Lemma 1. *An equilibrium profile (G,p) is Pareto efficient only if $supp(G)\cap(0,p) = \varnothing$.*

[1] For the usual reasons, we will break ties by assuming that the buyer purchases the good when indifferent.

[2] We adopt this notation from Roesler and Szentes (2017).

[3] An equilibrium profile is Pareto efficient if any deviations from the corresponding outcome necessarily reduce the buyer or seller surplus.

Proof. For contradiction, consider $\text{supp}(G) \cap (0, p) \neq \emptyset$, then we can construct a new cdf. The designer can provide additional information to all buyers with willingness to pay in $(0, p)$. For any type $\mu \in (0, p)$, the additional information induces a posterior valuation 0 with probability $1 - \alpha(\mu)$, a valuation p with probability $(1-\delta)\alpha(\mu)$, and valuation $p + \varepsilon$ with probability $\delta\alpha(\mu)$. The mean preserving constraint then requires $\alpha(\mu) = \frac{\mu}{p+\delta\varepsilon}$.

The proof is complete if there exists $\varepsilon, \delta > 0$ such that the seller still finds it optimal to set the price p. Let $\overline{\alpha} = \int_0^p \alpha(\mu) dG(\mu)$. We have increased the mass of buyers with valuation p by $(1-\delta)\overline{\alpha}$. The mass of buyers willing to buy at a price $p_1 \in (p, p+\varepsilon]$ is increased by $\delta\overline{\alpha}$. The seller surplus from charging a price above $p + \varepsilon$ is unchanged. As G was an outcome, we only need to verify whether the transformed demand creates incentives to price at some $p_1 \in (p, p+\varepsilon]$. Under the transformed demand, the seller's payoff from charging price p is

$$\pi + (p - c)\overline{\alpha}$$

The seller's payoff from charging a price $p_1 \in (p, p+\varepsilon]$ is bounded above by

$$\pi + (p + \varepsilon - c)\delta\overline{\alpha}$$

The profit maximizing price is p whenever $(p-c)\frac{1-\delta}{\delta} \geq \varepsilon > 0$, such an ε exists as by assumption G is non-degenerate and thus equilibrium price p is strictly greater than the cost c. Moreover, the construction in the proof above strictly increases both the buyer and seller surplus. $\square$

In general, many different market demands G can be feasible and lead to the same equilibrium price and outcome. These demands could be incomparable with respect to the convex order. Given a cost c, we will focus on the following class of demands determined by a market price p, a seller surplus π, and an expected valuation m.

$$H_p^\pi(s; m) = \begin{cases} 1 - \frac{\pi}{p-c} & s < p \\ 1 - \frac{\pi}{t(p,\pi,m)-c} & p \leq s < t(p,\pi,m) \\ 1 - \frac{\pi}{s-c} & t(p,\pi,m) \leq s < 1 \\ 1 & s = 1 \end{cases}$$

Where $t(p, \pi, m)$ is chosen such that the expectation of the distribution $H_p^\pi(\cdot; m)$ is indeed m. In particular, t is implicitly defined as the solution to the following whenever it exists

$$\frac{\pi}{1-c} + \int_t^1 s\frac{\pi}{(s-c)^2}ds + p\pi\left[\frac{1}{p-c} - \frac{1}{t-c}\right] = m \tag{1}$$

The solution to above exists for some $t \in [p, 1]$ whenever

$$\pi\left[\frac{p}{p-c} + \frac{1-p}{1-c}\right] \leq m \leq m^*(p, \pi) \tag{2}$$

Where

$$m^*(p,\pi) := \frac{\pi}{1-c} + \int_p^1 \pi\frac{s}{(s-c)^2}ds = \pi\left[\frac{p}{p-c} + \ln\frac{1-c}{p-c}\right]$$

In particular, a Pareto efficient distribution that induces price p, seller surplus π, and has the maximum possible expectation, $m^*(p,\pi)$, is such that $t(p,\pi,m) = p$.

If the first inequality in Equation 2 is not satisfied, then Equation 1 does not hold for any $t \in [p, 1]$. In this case we let $t = 1$ and consider the following three-point distribution for $H_p^\pi(\ ;m)$, the distribution H_p^π is described by three mass points $H_p^\pi(0) = 1 - \frac{\pi}{p-c}$, $\Delta(H_p^\pi, 1) = \frac{m - \pi\frac{p}{p-c}}{1-p}$, and $\Delta(H_p^\pi, p) = 1 - \Delta(H_p^\pi, 1) - H_p^\pi(0)$.

Whenever $m = \mathbf{E}F$ we will drop m from the notation and just write H_p^π and $t(p,\pi)$.

Define

$$\mathcal{H}(F) := \{G \in \mathcal{G}(F) \mid \exists\ \pi, p \text{ such that } G = H_p^\pi\}$$

These demands are iso-elastic (or unit elastic) on their support. For a given $H \in \mathcal{H}(F)$, the seller is indifferent between charging any positive price on the support of H. We will show that the above set of demands is sufficient to generate all possible Pareto efficient outcomes. Moreover, any Pareto efficient equilibrium demand can be transformed into a demand from $\mathcal{H}(F)$ through some information policy without affecting the welfare and price.

Lemma 2. *If (G, p) is a Pareto efficient equilibrium profile with corresponding outcomes (κ, π) then (H_p^π, p) is an outcome equivalent equilibrium profile. Moreover $H_p^\pi \succeq_{mps} G$.*

Proof. First we show the claim for a profile (G, p) with seller surplus π such that Equation 2 does not hold. Let $\widetilde{G}$ be a CDF such that $\Delta(\widetilde{G}, 1) = \int_p^1 \frac{s-p}{1-p}dG(s)$, $\Delta(\widetilde{G}, p) = \int_p^1 \frac{1-s}{1-p}dG(s)$, and $\widetilde{G}(0) = G(0)$. The CDF $\widetilde{G}$ is constructed by type-by-type garbling all types μ in the interval $[p, 1]$ into posteriors supported on $\{p, 1\}$,

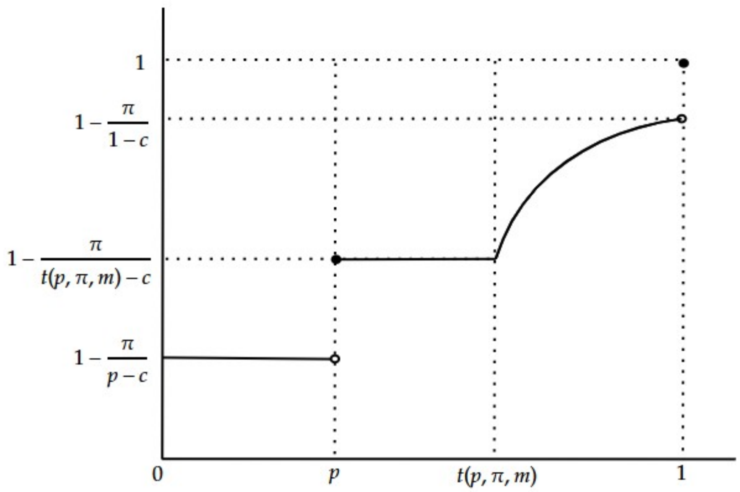

Figure 1: $H_p^\pi(s;m)$ when Equation 1 holds for $t \leq 1$.

thus $\widetilde{G} \succeq_{mps} G$. As Equation 2 does not hold it must be that $\Delta(\widetilde{G},1) \leq \frac{\pi}{1-c}$ and hence $p = \operatorname{argmax}_{[0,1]} (s-c)\cdot(1-\widetilde{G}(s)+\Delta(\widetilde{G},s))$. In particular, $(\widetilde{G},p)$ is an equilibrium profile with seller surplus π and satisfies $\widetilde{G} = H_p^\pi$.

Now let Equation 2 hold. As (G,p) is an equilibrium profile, we get that $p \in \operatorname{argmax}_s (s-c)\cdot(1-G(s)+\Delta(G,s))$. This implies that $G(s) \geq 1-\frac{\pi}{s-c}$ for all $s \in [p,1]$ and $\Delta(G,1) \leq \frac{\pi}{1-c}$. Thus the curve G lies weakly above the curve $H_p^\pi(\,\cdot\,; m^*(p,\pi))$. As (G,p) is an outcome we get that $\mathbf{E}G = \mathbf{E}F \leq m^*(p,\pi)$. By assumption Equation 2 is satisfied, thus there exists a unique $t(p,\pi) \in [p,1]$ which satisfies Equation 1 and H_p^π is well defined.

As both G and H_p^π are Pareto efficient demands for seller surplus π and price p we get that $G(0) = H_p^\pi(0) = 1-\frac{\pi}{p-c}$. Since $G(s) \geq 1-\frac{\pi}{s-c}$ for all $s \in [p,1]$, we get $\frac{\pi}{p-c} - \frac{\pi}{t(p,\pi)-c} \geq \Delta(G,p)$ as otherwise G lies strictly above H_p^π on $(p,1)$ which cannot happen as G and H_p^π have the same expectation. Moreover, as G lies weakly above $H_p^\pi(s) = 1-\frac{\pi}{s-c}$ on the interval $[t(p,\pi),1]$, Equation 1 (expectation is preserved) along with $G(0) = H_p^\pi(0)$ further imply that there exists $p_1 \in (p, t(p,\pi))$ such that $G(s) < H_p^\pi(s)$ for $s \in [p,p_1)$ and $G(s) \geq H_p^\pi(s)$ for $s \in [p_1, t(p,\pi))$. In particular, we have shown G crosses H_p^π once and from below.

The above, along with Equation 1 implies that $H_p^\pi \succeq_{mps} G$. As (G, p) is an equilibrium profile we get that $H_p^\pi \succeq_{mps} F$. By construction $p \in \text{argmax}_{s\in[0,1]} (s-c) \cdot (1 - H_p^\pi(s) + \Delta(H_p^\pi, s))$, thus (H_p^π, p) is an equilibrium profile. It induces the same seller surplus π, price p, probability of trade $1 - \Delta(G, 0)$, and buyer surplus. $\square$

Main Results

For our main results, we will first show comparative statics between equilibrium prices and the informativeness of the buyer's information policy. Using this, we characterize the Pareto-efficient and buyer-optimal demands when $c \geq 0$. The comparative static result also allows us to identify the maximum and minimum prices that can implement a given equilibrium seller surplus.

Before stating the theorem, we will define these extremal prices for some equilibrium seller surplus π. The highest price that implements π is defined implicitly as the solution to the following

$$\overline{p}_\pi = t(\overline{p}_\pi, \pi)$$

The lowest price that implements π is defined implicitly as a solution to the following

$$\int_0^{\underline{p}_\pi} \left(H_{\underline{p}_\pi}^\pi(s) - F(s) \right) ds = 0$$

As H_p^π is constant and equal to $1 - \frac{\pi}{p-c}$ for $s \leq p$ we get $\int_0^{\underline{p}_\pi} \left(H_{\underline{p}_\pi}^\pi(s) - F(s) \right) ds = 0$ implies that $c < \underline{p}_\pi$. Moreover the lowest price $\underline{p}_\pi$ equals the highest price $\overline{p}_\pi$ only if F is constant and equal to $1 - \frac{\pi}{p-c}$ on $[0, \overline{p}_\pi]$.

We are now in a position to state the main result of the paper. Consider the digital platform discussed in the introduction. The platform can provide the buyer with more or less information at the interim stage. A regulator may be concerned about high prices and its consequences for welfare. Yet high prices are a direct consequence of improved information provided by the platform and the resulting spread in buyer willingness to pay. Indeed, there exists a range of prices implemented by a correspondingly ordered range of information policies without any impact on producer welfare. The implications for buyer welfare are addressed in the next section.

Theorem 1. *Suppose that $c \geq 0$. For any $p_0 < p_1$ and π, if $(H_{p_0}^\pi, p_0)$ and $(H_{p_1}^\pi, p_1)$ are equilibrium profiles then $H_{p_1}^\pi \succeq_{mps} H_{p_0}^\pi$. Moreover, if (G, p) is a Pareto efficient*

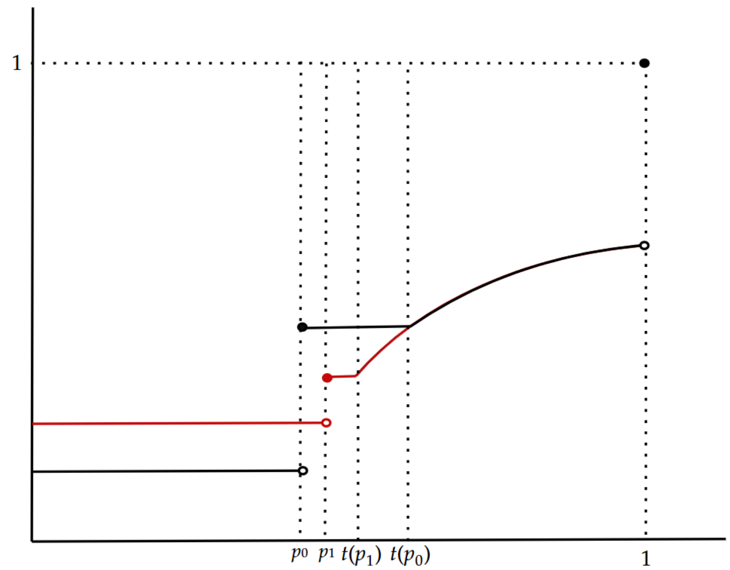

Figure 2: Single crossing between $H^{\pi}_{p_0}$ in black and $H^{\pi}_{p_1}$ in red.

equilibrium profile with seller surplus π then $\underline{p}_{\pi} \leq p \leq \overline{p}_{\pi}$ and for any price s such that $\underline{p}_{\pi} \leq s \leq \overline{p}_{\pi}$, the profile (H^{π}_s, s) is an equilibrium profile.

Proof. By definition, to prove $H^{\pi}_{p_1} \succeq_{mps} H^{\pi}_{p_0}$ it suffices to show that $t(p_0, \pi) \geq t(p_1, \pi)$. By Equation 1 we get that

$$p_0 \left(\frac{\pi}{p_0 - c} - \frac{\pi}{t(p_0, \pi) - c} \right) + \int_{t(p_0,\pi)}^{1} s \frac{\pi}{(s-c)^2} \, ds$$
$$= p_1 \left(\frac{\pi}{p_1 - c} - \frac{\pi}{t(p_1, \pi) - c} \right) + \int_{t(p_1,\pi)}^{1} s \frac{\pi}{(s-c)^2} \, ds$$

After rearranging we get

$$p_0 \left(\frac{\pi}{p_0 - c} - \frac{\pi}{p_1 - c} \right) = (p_1 - p_0) \left(\frac{\pi}{p_1 - c} - \frac{\pi}{t(p_1, \pi) - c} \right)$$
$$+ \int_{t(p_1,\pi)}^{t(p_0,\pi)} \pi \left(\frac{s}{(s-c)^2} - p_0 \frac{1}{(s-c)^2} \right) ds$$

implying

$$\frac{p_1 - p_0}{p_1 - c}\left[\frac{c(t(p_1,\pi) - p_0) + p_1(p_0 - c)}{(t(p_1,\pi) - c)(p_0 - c)}\right] = \int_{t(p_1,\pi)}^{t(p_0,\pi)} \left(\frac{s}{(s-c)^2} - p_0\frac{1}{(s-c)^2}\right)\, ds$$

As $t(p_1,\pi), t(p_0,\pi) \geq p_0 \geq c \geq 0$ we get that the above equation holds only if $t(p_0,\pi) \geq t(p_1,\pi)$ (See Figure 2). This establishes the first claim.

For the next claim, recall that by Lemma 2 if (G,p) is a Pareto-efficient equilibrium profile then (H^π_p, p) is an outcome-equivalent equilibrium profile. If $p < t(p,\pi)$, then consider some $\varepsilon > 0$ such that $p+\varepsilon = t(p+\varepsilon,\pi)$. By the previous claim, we get that $t(p,\pi) > t(s,\pi)$ for all $s > p$ whenever t is well defined, thus an $\varepsilon > 0$ that satisfies the required equality exists. Finally, note that any price $s > \overline{p}_\pi$ cannot be supported by an efficient equilibrium profile with seller surplus π as Equation 1 cannot be satisfied.

For the lowest price $\underline{p}_\pi$, note that by Lemma 2 if (G,p) is a Pareto-efficient equilibrium profile then (H^π_p, p) is an outcome-equivalent equilibrium profile. As (H^π_p, p) is an equilibrium profile we get that $H^\pi_p \succeq_{mps} F$ or equivalently $\int_0^s H^\pi_p(\mu) - F(\mu)d\mu \geq 0$ for all s. As H^π_p is constant and F non-decreasing on $[0,p)$, if $\int_0^p H^\pi_p(\mu) - F(\mu)d\mu > 0$ then $\int_0^s H^\pi_p(\mu) - F(\mu)d\mu > 0$ for all $s \leq p$. We can choose an $\varepsilon > 0$ such that $\int_0^s H^\pi_{p-\varepsilon}(\mu) - F(\mu)d\mu > 0$ for all $s \leq p - \varepsilon$. By construction we get that $H^\pi_{p-\varepsilon}(s) \geq H^\pi_p(s)$ for all $s \geq p-\varepsilon$, combining this with the assumption that $H^\pi_p \succeq_{mps} F$ we obtain that $H^\pi_{p-\varepsilon} \succeq_{mps} F$. Thus, $(H^\pi_{p-\varepsilon}, p-\varepsilon)$ is an equilibrium profile. We can then choose ε large enough such that $\int_0^{p-\varepsilon} H^\pi_{p-\varepsilon}(\mu) - F(\mu)d\mu = 0$, in particular $p - \varepsilon = \underline{p}_\pi$. Moreover, any price $p' < \underline{p}_\pi$ is such that $\int_0^{p'} H^\pi_{p'}(\mu) - F(\mu)d\mu < 0$ which implies that $(H^\pi_{p'}, p')$ is not an equilibrium profile.

To complete the proof, note that we have shown any seller surplus π that is implemented by some Pareto efficient equilibrium profile (G,p) must be such that $\underline{p}_\pi \leq p \leq \overline{p}_\pi$. Additionally, $(H^\pi_{\underline{p}_\pi}, \underline{p}_\pi)$ and $(H^\pi_{\overline{p}_\pi}, \overline{p}_\pi)$ are equilibrium profiles which implies that $H^\pi_{\underline{p}_\pi} \succeq_{mps} F$. By the first claim of the theorem, for any $s \in [\underline{p}_\pi, \overline{p}_\pi]$ the isoelastic demand $H^\pi_{\overline{p}_\pi} \succeq_{mps} H^\pi_s \succeq_{mps} H^\pi_{\underline{p}_\pi}$. □

Theorem 1 shows that for a fixed seller surplus, more informed buyers face higher equilibrium prices. Additionally, Theorem 1 uses this relationship between prices and informativeness to identify the highest and the smallest equilibrium prices

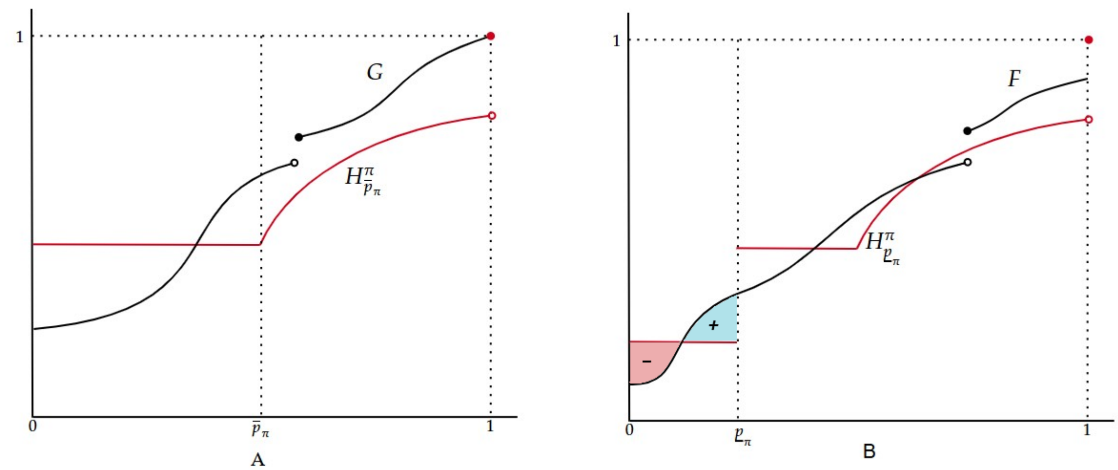

Figure 3: On the left (A) we have single crossing between the most informative distribution $H^{\pi}_{\bar{p}_\pi}$ in red and some generic equilibrium demand G in black. On the right (B) we have the least informative distribution $H^{\pi}_{\underline{p}_\pi}$ in red and the interim demand F in black. The area under the region '-' in red and the region '+' in blue are equal by definition of $\underline{p}_\pi$.

that support seller surplus π ; these in turn correspond to the most and the least informative Pareto efficient demands that induce a given seller surplus π respectively. (see Figure 3)

Welfare

In this section we use the distributions $H^{\pi}_{\underline{p}_\pi}$ to characterize the Pareto efficient welfare. We show that high prices can be a consequence of valuable information rather than a source of consumer harm: holding seller surplus fixed, the more informative disclosures associated with higher prices leave buyer surplus unchanged in the no-gap case and strictly increase it when $c > 0$. Coupled with Theorem 1, this shows that total welfare is invariant to price in the no-gap case and strictly increasing in price when $c > 0$, as platforms provide buyers with more information. We also identify the buyer-optimal outcome. We also use the buyer-optimal outcome to characterize all possible Pareto-efficient equilibrium seller surpluses. We start by describing the set of Pareto-efficient seller surpluses, taking the buyer-optimal outcome as given.

Proposition 1. *Suppose $c \geq 0$. If (π^*, κ^*) is the buyer-optimal equilibrium outcome then $(H^{\pi}_{\overline{p}_\pi}, \overline{p}_\pi)$ is an equilibrium profile for all $\pi^* \leq \pi \leq (1-c)\mathbf{E}F$. In particular any seller surplus $\pi \in [\pi^*, (1-c)\mathbf{E}F]$ can be achieved by some Pareto efficient equilibrium profile.*

Proof. To prove the claim, we will first show that $\overline{p}_\pi$ is increasing in π. Note that $\overline{p}_\pi = t(\overline{p}_\pi, \pi)$ then by Equation 1

$$\frac{1}{1-c} + \int_{\overline{p}_\pi}^{1} \frac{s}{(s-c)^2}\, ds = \frac{\mathbf{E}F}{\pi}$$

The right-hand side above is decreasing in π; this implies the left-hand side is decreasing in π, and hence, $\overline{p}_\pi$ is increasing in π. Assume that (G, p) is a buyer-optimal equilibrium profile with outcome (π^*, κ^*). By Theorem 1 $(H^{\pi^*}_{\overline{p}_{\pi^*}}, \overline{p}_{\pi^*})$ is an equilibrium profile and thus $H^{\pi^*}_{\overline{p}_{\pi^*}} \succeq_{mps} F$. Consider some $\pi \in (\pi^*, 1]$, then $\overline{p}_\pi > \overline{p}_{\pi^*}$. By definition of $H^{\pi}_{\overline{p}_\pi}$ and $H^{\pi^*}_{\overline{p}_{\pi^*}}$, we get that $\Delta(H^{\pi^*}_{\overline{p}_{\pi^*}}, 0) < \Delta(H^{\pi}_{\overline{p}_\pi}, 0)$, $\mathbf{E}H^{\pi^*}_{\overline{p}_{\pi^*}} = \mathbf{E}H^{\pi}_{\overline{p}_\pi}$ and that $\Delta(H^{\pi}_{\overline{p}_\pi}, 0)$ is constant until $\overline{p}_\pi$. Thus, $H^{\pi^*}_{\overline{p}_{\pi^*}}$ crosses $H^{\pi}_{\overline{p}_\pi}$ once and from below. Putting together this implies $H^{\pi}_{\overline{p}_\pi} \succeq_{mps} H^{\pi^*}_{\overline{p}_{\pi^*}} \succeq_{mps} F$, thus $(H^{\pi}_{\overline{p}_\pi}, \overline{p}_\pi)$ is an equilibrium profile. To complete the proof, note that as $c \geq 0$ the highest possible seller surplus is $(1-c)\mathbf{E}F$. It is achieved by a fully informative policy with a price of 1.

The above arguments establish the feasibility of $H^{\pi}_{\overline{p}_\pi}$ for all $\pi \in [\pi^*, (1-c)\mathbf{E}F]$, for Pareto efficiency of $H^{\pi}_{\overline{p}_\pi}$ it suffices to note that $c \geq 0$ implies that the buyer surplus is increasing in price (for fixed seller surplus). $\square$

Proposition 1 characterizes the whole set of Pareto-efficient seller surpluses by showing that a Pareto-efficient equilibrium profile with seller surplus π can be transformed, via targeted disclosure, into a Pareto-efficient equilibrium profile with higher seller surplus. The lowest achievable seller surplus corresponds to the buyer-optimal outcome, while the highest seller surplus corresponds to an information policy that provides complete information to the buyer.

Now we will shift focus to buyer-optimal outcomes. To that end, note that the buyer surplus against the demand H^{π}_{p} is given by

$$\int_{p}^{1} (s-p)\, dH^{\pi}_{p}(s) = \mathbf{E}F - p\frac{\pi}{p-c} \tag{3}$$

Differentiating the right-hand side of Equation 3 with respect to p we get

$$\frac{c\pi}{(p-c)^2}$$

When $c = 0$, the no-gap case, price is neutral with respect to buyer welfare holding seller surplus π fixed. To understand this, recall from Theorem 1, price increases starting from a Pareto efficient outcome are the consequence of improved information. In the no-gap case the welfare costs of such price increases are exactly offset by the value of improved information and the net effect is that buyer surplus is constant. Indeed it follows from Theorem 1 that across the range $[\underline{p}_\pi, \overline{p}_\pi]$, price variation is completely welfare neutral starting from a Pareto efficient outcome with seller surplus π.

When $c > 0$, buyer surplus is increasing in price; hence, holding seller surplus π fixed, the buyer strictly prefers higher prices. Therefore, if the seller surplus π is achieved on the Pareto frontier, then it is achieved by the most informative distribution $H^{\pi}_{\overline{p}_\pi}$. In particular, by Theorem 1, when $c \geq 0$, the seller surplus corresponding to the buyer-optimal equilibrium outcome solves the following optimization problem

$$\inf_{\pi \in [0,1]} \pi \frac{\overline{p}_\pi}{\overline{p}_\pi - c} \quad \text{subject to} \quad \int_0^x \left(H^{\pi}_{\overline{p}_\pi}(s) - F(s) \right) \, ds \geq 0 \quad \text{for all } x \in [0,1]$$

Constructing a Buyer-Optimal Information Policy

We now specialize to the no-gap case and select the most informative Pareto-efficient implementation of the buyer-optimal outcome. We first characterize the demand and seller surplus supporting this outcome and then, for regular priors and under an additional monotonicity condition, explicitly construct the underlying type-dependent information policy.

In the no-gap case, Equation 1 and the definition of $\overline{p}_\pi$ imply that $\overline{p}_\pi = \exp\left(1 - \frac{\mathbf{E}F}{\pi}\right)$. Combining this with the optimization problem from before, we recover the following simplified optimization problem for the buyer-optimal outcome in the no-gap case.

$$\inf_{\pi \in [0,1]} \pi \quad \text{subject to} \quad \int_0^x \left(H^{\pi}_{\exp\left(1 - \frac{\mathbf{E}F}{\pi}\right)}(s) - F(s) \right) \, ds \geq 0 \quad \text{for all } x \in [0,1] \tag{4}$$

Using this formulation, we can calculate the buyer-optimal equilibrium outcome and construct an information policy that implements this outcome. Before presenting this construction we highlight a useful necessary condition for $H^{\pi}_{\overline{p}_{\pi}}$ to be buyer-optimal.

Lemma 3. *Suppose $c = 0$ (no-gap) and $\Delta(F, 1) = 0$. If $(H^{\pi^*}_{\overline{p}_{\pi^*}}, \overline{p}_{\pi^*})$ is the buyer-optimal outcome then there exists a type $\hat{\mu} \in [\overline{p}_{\pi^*}, 1)$ such that*

$$\int_0^{\hat{\mu}} H^{\pi^*}_{\overline{p}_{\pi^*}}(s) - F(s)\, ds = 0 \tag{5}$$

and

$$H^{\pi^*}_{\overline{p}_{\pi^*}}(\hat{\mu}) = F(\hat{\mu}) \tag{6}$$

Proof. Assume for contradiction that Equation 5 does not hold. As $H^{\pi^*}_{\overline{p}_{\pi^*}} \succeq_{mps} F$ it follows that $\int_0^{\mu} H^{\pi^*}_{\overline{p}_{\pi^*}}(s) - F(s)\, ds > 0$ for all $\mu \in [\overline{p}_{\pi^*}, 1)$. Moreover, as $H^{\pi^*}_{\overline{p}_{\pi^*}}$ is constant on $(0, \overline{p}_{\pi^*}]$ and by right continuity of F, $H^{\pi^*}_{\overline{p}_{\pi^*}} \succeq_{mps} F$ implies that $\int_0^{\mu} H^{\pi^*}_{\overline{p}_{\pi^*}}(s) - F(s)\, ds > 0$ for all $\mu \in (0, \overline{p}_{\pi^*}]$. In particular, $H^{\pi^*}_{\overline{p}_{\pi^*}}(0) - F(0) > 0$.

For $\mu \in (0, 1)$ define the normalized slack

$$\Psi_{\pi^*}(\mu) := \frac{1}{\mu(1-\mu)} \int_0^{\mu} \left(H^{\pi^*}_{\overline{p}_{\pi^*}}(s) - F(s) \right) ds.$$

This function extends continuously to $[0, 1]$ with endpoints given by $\lim_{\mu \downarrow 0} \Psi_{\pi^*}(\mu) = 1 - \frac{\pi^*}{\overline{p}_{\pi^*}} - F(0) > 0$ and

$$\lim_{\mu \uparrow 1} \Psi_{\pi^*}(\mu) = \lim_{\mu \uparrow 1} \frac{\int_{\mu}^{1} \left(F(s) - H^{\pi^*}_{\overline{p}_{\pi^*}}(s) \right) ds}{\mu(1-\mu)} = \Delta(H^{\pi^*}_{\overline{p}_{\pi^*}}, 1) - \Delta(F, 1)$$

By assumption $\Delta(F, 1) = 0$ and the difference $\Delta(H^{\pi^*}_{\overline{p}_{\pi^*}}, 1) - \Delta(F, 1) > 0$. In particular, $\min_{\mu \in [0,1]} \Psi_{\pi^*}(\mu) > 0$. Consider the difference $\Delta_{\varepsilon}\Psi(\mu) = \Psi_{\pi^*}(\mu) - \Psi_{\pi^* - \varepsilon}(\mu)$. By definition of $H^{\pi}_{\overline{p}_{\pi}}$ we get that $\Delta_{\varepsilon}\Psi(\mu) < \frac{1}{1 - \overline{p}_{\pi^*}} \left(\frac{\pi^* - \varepsilon}{\overline{p}_{\pi^* - \varepsilon}} - \frac{\pi^*}{\overline{p}_{\pi^*}} \right)$ for $\mu \leq \overline{p}_{\pi^*}$ and $\Delta_{\varepsilon}\Psi(\mu) < \frac{\varepsilon}{\overline{p}_{\pi^*}}$ for $\mu \geq \overline{p}_{\pi^*}$. For a small enough ε, $\Psi_{\pi^* - \varepsilon}(\mu) \geq \min_{\mu \in [0,1]} \Psi_{\pi^*}(\mu) - \Delta_{\varepsilon}\Psi(\mu) > 0$. Thus $\Psi_{\pi^* - \varepsilon}(\mu) > 0$ for all $\mu \in (0, 1)$ which implies $\int_0^{\mu} H^{\pi^* - \varepsilon}_{\overline{p}_{\pi^* - \varepsilon}}(s) - F(s) ds > 0$ for all $\mu \in (0, 1)$. This contradicts the optimality of π^*.

From above $\int_0^{\mu} H_{\overline{p}_{\pi^*}}^{\pi^*}(s) - F(s)\, ds \geq 0$ and $\int_0^{\hat{\mu}} H_{\overline{p}_{\pi^*}}^{\pi^*}(s) - F(s)\, ds = 0$, thus we get that on a right neighborhood of $\hat{\mu}$ the prior $F(\mu) \leq H_{\overline{p}_{\pi^*}}^{\pi^*}(\mu)$. Similarly, on a left neighborhood of $\hat{\mu}$ we have $F(\mu) \geq H_{\overline{p}_{\pi^*}}^{\pi^*}(\mu)$. Thus, by continuity of $H_{\overline{p}_{\pi^*}}^{\pi^*}$, the prior F is continuous at $\hat{\mu}$ and $H_{\overline{p}_{\pi^*}}^{\pi^*}(\hat{\mu}) = F(\hat{\mu})$. □

Now we can state the main result of this section, which constructs the buyer-optimal equilibrium profile and an implementing information policy for a regular prior F. A prior F is *regular* whenever it admits a positive density $f > 0$ and

$$\mu - \frac{1 - F(\mu)}{f(\mu)} \quad \text{is strictly increasing in } \mu$$

Theorem 2 shows that a regular prior implies that the necessary conditions in Lemma 3 are also sufficient to solve for the buyer-optimal outcome. The following result uses these necessary (and sufficient) conditions, Equation 5 and Equation 6, to construct an information policy that implements the buyer-optimal outcome.

Theorem 2. *Suppose that $c = 0$ (no-gap) and F is regular. A seller surplus π^* is the solution to Equation 4, i.e. π^* is buyer-optimal, if and only if there exists a type $\hat{\mu} \in [\overline{p}_{\pi^*}, 1)$ such that Equation 5 and Equation 6 hold.*

In addition, if $\mu^2 f(\mu)$ is non-decreasing on $[\hat{\mu}, 1]$, then the buyer-optimal outcome is implemented by the following information policy

$$\begin{pmatrix} \rho_1(\mu) \\ \rho_0(\mu) \end{pmatrix} = \begin{cases} \begin{pmatrix} 0 & 1 \\ \frac{q(\mu)-\mu}{(1-\mu)q(\mu)} & \frac{\mu(1-q(\mu))}{(1-\mu)q(\mu)} \end{pmatrix} & \mu \leq \hat{\mu} \\[2em] \begin{pmatrix} \frac{(1-\mu)q(\mu)}{\mu(1-q(\mu))} & \frac{\mu-q(\mu)}{\mu(1-q(\mu))} \\ 1 & 0 \end{pmatrix} & \mu > \hat{\mu} \end{cases}$$

Where $q(\mu)$ is defined implicitly as the solution to the following

$$q(\mu) = \overline{p}_{\pi^*} \exp\left(\frac{1}{\pi^*} \int_0^{\mu} s dF(s)\right) \quad \textit{if } \mu \leq \hat{\mu}$$

and

$$\pi^* \int_{\hat{\mu}}^{q(\mu)} \frac{1-s}{s^2} ds \; = \; \int_{\hat{\mu}}^{\mu} (1-s) dF(s) \quad \textit{if } \mu > \hat{\mu}$$

Proof. The necessity of Equation 6 and Equation 5 follows from Lemma 3. For sufficiency, assume that $(\hat{\mu}, \pi)$ satisfy Equation 6 and Equation 5. In particular Equation 6 implies $\pi = \hat{\mu}(1 - F(\hat{\mu}))$. Substituting this and $\overline{p}_\pi = \exp\left(1 - \frac{\mathbf{E}F}{\pi}\right)$ into Equation 5 we get

$$\int_0^{\hat{\mu}} H^{\pi}_{\overline{p}_\pi}(s) - F(s)\, ds = \hat{\mu} - \mathbf{E}F + \pi \ln \frac{1}{\hat{\mu}} - \int_0^{\hat{\mu}} F(s)\, ds$$

Differentiating the right hand side above with respect to $\hat{\mu}$ yields $1 - \frac{\pi}{\hat{\mu}} - \frac{\partial \pi}{\partial \hat{\mu}} \ln \hat{\mu} - F(\hat{\mu})$. After substituting $\pi = \hat{\mu}(1 - F(\hat{\mu}))$ we obtain $f(\hat{\mu}) \ln \hat{\mu} \left(\hat{\mu} - \frac{1-F(\hat{\mu})}{f(\hat{\mu})}\right)$. For regular prior this derivative changes sign once from positive to negative. The expression, $\int_0^1 H^{\pi}_{\overline{p}_\pi}(s) - F(s)\, ds$ is 0, implying that $\hat{\mu}$ is the unique point, less than 1, which solves Equation 5 and Equation 6. Moreover $\hat{\mu} - \frac{1-F(\hat{\mu})}{f(\hat{\mu})} \leq 0$.

An optimal solution, π^* to Equation 4 exists and by Lemma 3 there exists $\mu \in (\overline{p}_{\pi^*}, 1)$ such that Equation 6 and Equation 5 hold. By the uniqueness argument for regular prior we get $\pi = \pi^*$.

To verify that ρ implements the buyer-optimal outcome π^* we will first show that ρ is feasible and well defined. The information policy corresponds to a binary support posterior distribution for each type, assigning posteriors 0 and $q(\mu)$, with $\mu \leq q(\mu) \leq \hat{\mu}$, to types $\mu \leq \hat{\mu}$ and posteriors 1 and $q(\mu)$, with $\hat{\mu} \leq q(\mu) \leq \mu$, to $\mu > \hat{\mu}$.

For $\mu \leq \hat{\mu}$ we have that $q(\mu) = \overline{p}_{\pi^*} \exp\left(\frac{1}{\pi^*} \int_0^{\mu} s dF(s)\right)$, this is strictly increasing in μ with $q(0) = \overline{p}_{\pi^*}$. By Equation 5, $q(\hat{\mu}) = \overline{p}_{\pi^*} \exp\left(\frac{1}{\pi^*} \int_0^{\hat{\mu}} s dF(s)\right) = \overline{p}_{\pi^*} \exp\left(\frac{1}{\pi^*} \int_0^{\hat{\mu}} s dH^{\pi^*}_{\overline{p}_{\pi^*}}(s)\right)$. Using the definition of $\overline{p}_{\pi^*}$ we get

$$q(\hat{\mu}) = \exp\left(1 - \frac{\pi^* - \int_{\hat{\mu}}^1 s dH^{\pi^*}_{\overline{p}_{\pi^*}}(s)}{\pi^*}\right) = \hat{\mu}$$

Now note that $\frac{d}{d\mu} \frac{q(\mu)}{\mu} = q(\mu) \left(\frac{f(\mu)}{\pi^*} - \frac{1}{\mu^2}\right)$, by Equation 6 it follows

$$\frac{d}{d\mu} \frac{q(\mu)}{\mu} = q(\mu) \left(\frac{f(\mu)}{\hat{\mu}(1 - F(\hat{\mu}))} - \frac{1}{\mu^2}\right)$$

Recall from before that Equation 5 and Equation 6 imply that $\hat{\mu} - \frac{1-F(\hat{\mu})}{f(\hat{\mu})} \leq 0$, by regularity this implies $\mu - \frac{1-F(\mu)}{f(\mu)} \leq 0$ and $\mu(1-F(\mu)) \leq \hat{\mu}(1-F(\hat{\mu}))$ for $\mu \leq \hat{\mu}$. In particular $\frac{d}{d\mu}\frac{q(\mu)}{\mu} \leq q(\mu)\left(\frac{f(\mu)}{\mu(1-F(\mu))} - \frac{1}{\mu^2}\right) = q(\mu)\frac{f(\mu)}{\mu^2(1-F(\mu))}\left(\mu - \frac{1-F(\mu)}{f(\mu)}\right)$.

Thus, for $\mu \leq \hat{\mu}$ we get $\frac{d}{d\mu}\frac{q(\mu)}{\mu} \leq 0$. As $q(\hat{\mu}) = \hat{\mu}$ this implies that $q(\mu) \geq \mu$ for all $\mu \leq \hat{\mu}$.

Now we verify the feasibility of ρ for $\mu > \hat{\mu}$. The posterior $q(\mu)$ solves the equation

$$\pi^* \int_{\hat{\mu}}^{q(\mu)} \frac{1-s}{s^2} ds \;=\; \int_{\hat{\mu}}^{\mu} (1-s) dF(s) \quad \text{if} \;\; \mu > \hat{\mu}$$

The right-hand side above is increasing in μ and is zero at $\mu = \hat{\mu}$, to match this $q(\mu)$ is also increasing in μ and $\lim_{\mu \downarrow \hat{\mu}} q(\mu) = \hat{\mu}$. By assumption $\mu^2 f(\mu)$ is non-decreasing thus $\int_{\hat{\mu}}^{\mu} (1-s)\left(\frac{\pi}{s^2} - f(s)\right) ds$ is single peaked. Moreover, by Equation 5 and Equation 6 we get $\int_{\hat{\mu}}^{1} (1-s)\left(\frac{\pi^*}{s^2} - f(s)\right) ds = \int_{\hat{\mu}}^{1} (1-s)\frac{\pi^*}{s^2} ds - \int_{\hat{\mu}}^{1} (1-s) dH^{\pi^*}_{\overline{p}_{\pi^*}} = 0$. Thus, $\int_{\hat{\mu}}^{\mu} (1-s)\left(\frac{\pi}{s^2} - f(s)\right) ds > 0$ which implies $q(\mu) \leq \mu$. This establishes that ρ is a well-defined and feasible information policy.

Now we can check whether ρ implements $H^{\pi^*}_{\overline{p}_{\pi^*}}$. First note $\mathbf{E}_\rho\{\nu = 0\} = \int_0^{\hat{\mu}} 1 - \frac{\mu}{q(\mu)} dF(\mu) = F(\hat{\mu}) - \int_0^{\hat{\mu}} \mu f(\mu) \exp\left(\frac{1}{\pi^*}\int_\mu^1 s dF(s) - 1\right) d\mu = F(\hat{\mu}) + \frac{\pi^*}{q(\hat{\mu})} - \frac{\pi^*}{\overline{p}_{\pi^*}} = 1 - \frac{\pi^*}{\overline{p}_{\pi^*}} = \Delta(H^{\pi^*}_{\overline{p}_{\pi^*}}, 0)$. For $\mu \in [\overline{p}_{\pi^*}, \hat{\mu}]$, the density of the market demand induced by ρ at μ is given by $\frac{d}{d\mu}\mathbf{E}_\rho\{\nu \leq \mu\} = \frac{d}{d\mu}\int_0^{q^{-1}(\mu)} \frac{s}{q(s)} dF(s) = \frac{d}{d\mu}\pi^*\left(\frac{1}{\overline{p}_{\pi^*}} - \frac{1}{q(q^{-1}(\mu))}\right) = \frac{\pi^*}{\mu^2} = dH^{\pi^*}_{\overline{p}_{\pi^*}}(\mu)$. Now consider $\mu \in (\hat{\mu}, 1)$, the density of the market demand induced by ρ at μ is given by $\frac{d}{d\mu}\mathbf{E}_\rho\{\nu \leq \mu\} = \frac{d}{d\mu}\int_{\hat{\mu}}^{q^{-1}(\mu)} \frac{1-s}{1-q(s)} dF(s)$, using change of variable $\mu = q(s)$ we get that $d\mu = q'(s)ds = \frac{(1-s)q^2(\mu)}{\pi^*(1-q(s))} f(s) ds$. Thus $\frac{d}{d\mu}\mathbf{E}_\rho\{\nu \leq \mu\} = \frac{d}{d\mu}\int_{\hat{\mu}}^{\mu} \frac{\pi^*}{s^2} ds = dH^{\pi^*}_{\overline{p}_{\pi^*}}(\mu)$. This completes the proof. □

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